

6.2.3 Nonuniform Quantization

- The SNR is an indication of the quality of the received signal. Ideally the SNR should be constant (the same quality) for all values of the signal power.
- From Eq. (6.34), it can be seen that the SNR is directly proportional to the signal power $S_0 = \overline{m^2(t)}$,
- The signal power varies from speaker to speaker by as much as 40 dB (a power ratio of 10^4) and varies also with the lengths of the connecting circuits.
- Even for the same speaker, the quality of the received signal will deteriorate markedly when the person speaks softly.
- Statistically, it is found that smaller amplitudes predominate in speech and larger amplitudes are much less frequent.

- This means the SNR will be low most of the time.
- Since the quantization noise N_q is directly proportional to the square of the step size Δv [$N_q = (\Delta v)^2/2$].
- So, the SNR can be fixed if we make N_q varies in accordance with S_0 . This is achieved by using smaller Δv for small signal amplitudes and larger Δv for high signal amplitudes.
- This technique is called **nonuniform quantizing**, as shown in Fig. 6.15a.

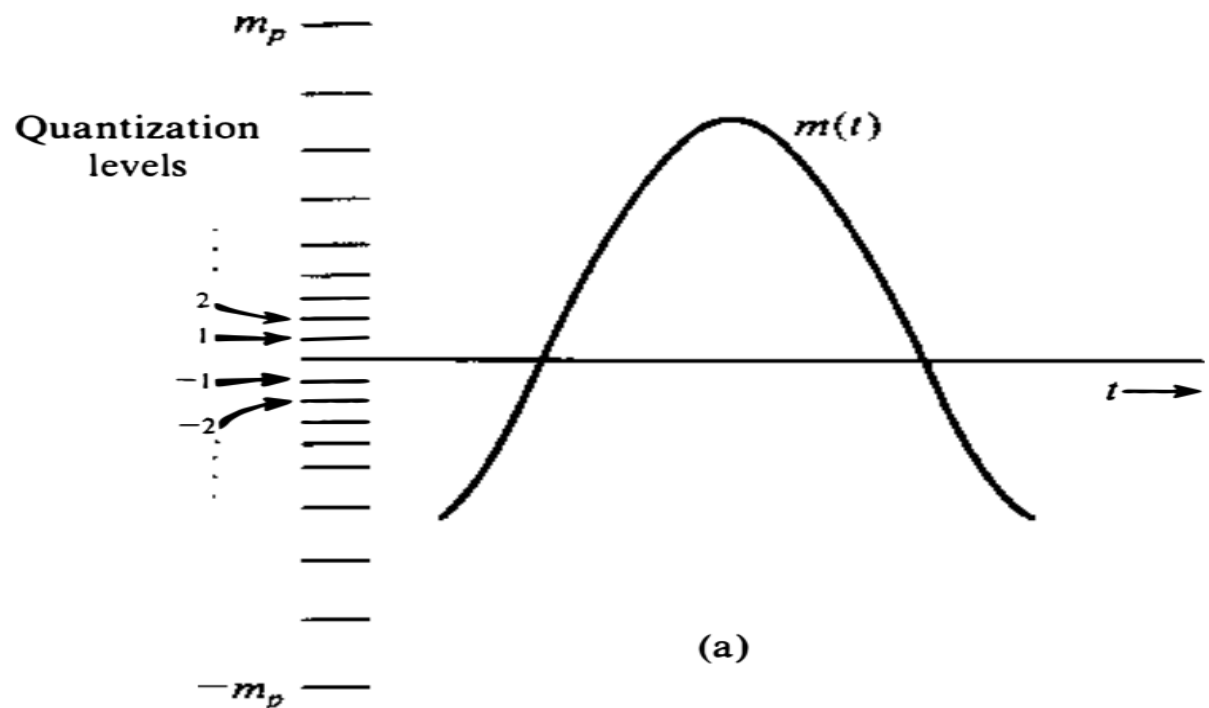
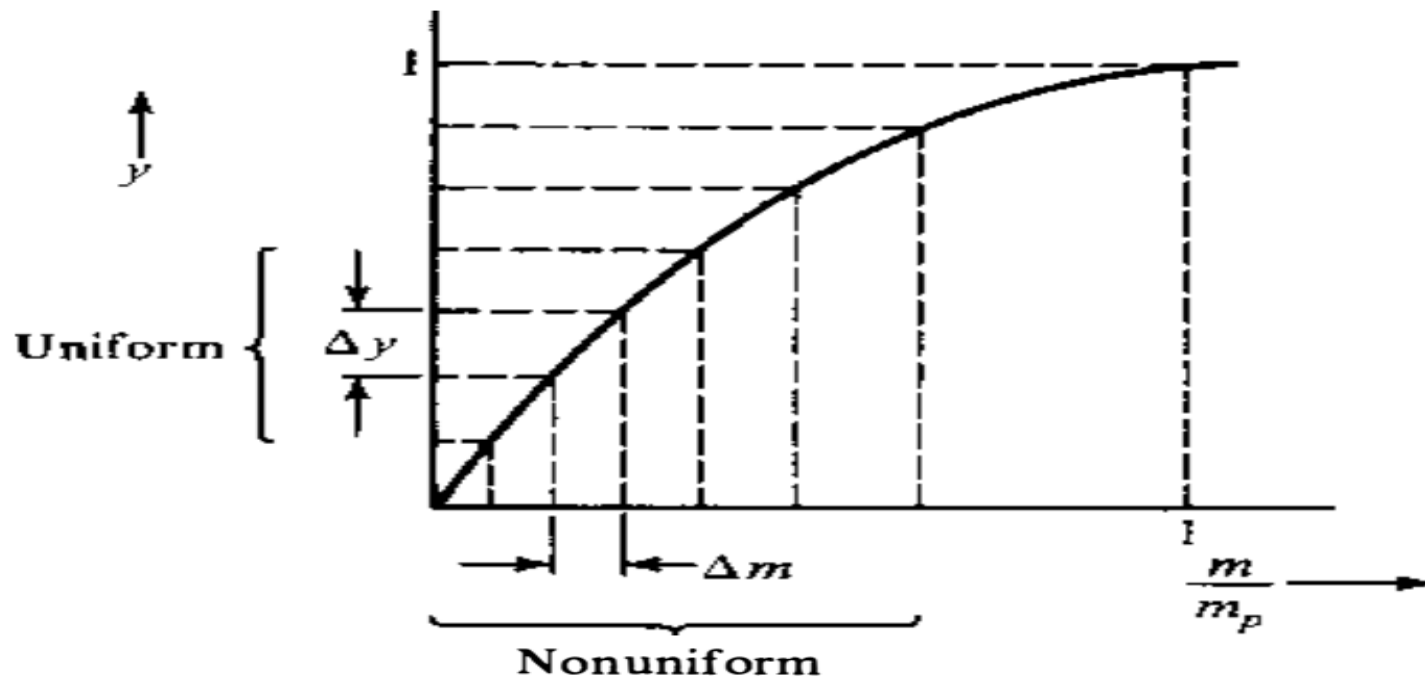


Figure 6.15:
Nonuniform
quantization

(a)

The same result is obtained by first compressing signal samples and then using a uniform quantization. The input-output characteristics of the compressor are shown in Fig. 6.15b. The horizontal axis is the normalized input signal (i.e., the input signal amplitude m divided by the signal peak value m_p). The vertical axis is the output signal of the quantizer y which represents the level magnitude ($y_1 = L_1, y_2 = L_2 \dots$)



(b)

- There are two compression laws that have been accepted as desirable standards by the International Telecommunication Union- Telecommunication sector (ITU-T): the **μ-law** used in North America and Japan, and the **A-law** used in Europe and the rest of the world and on international routes.
- Both the μ-law and the A-law curves have odd symmetry about the vertical axis. The μ-law (for positive amplitudes) is given by:

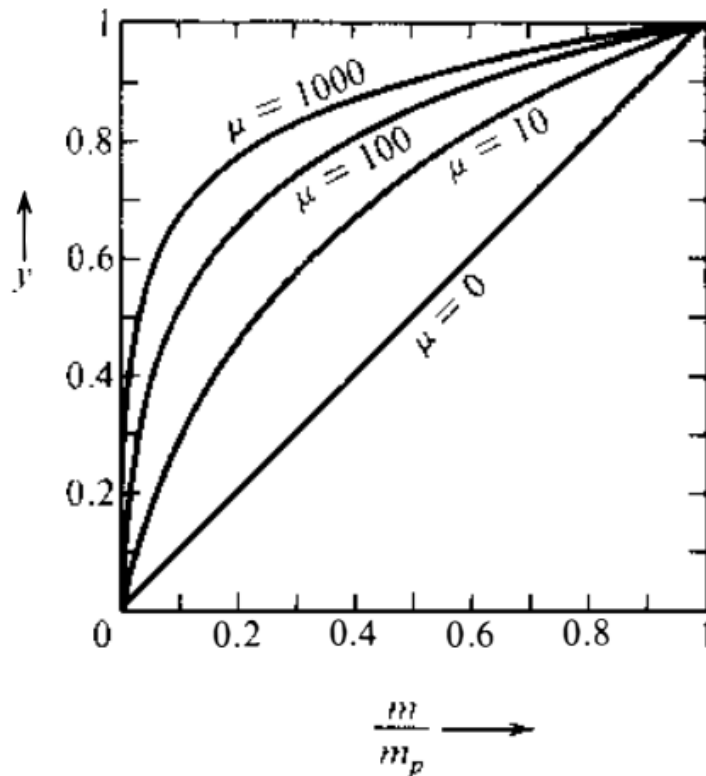
$$y = \frac{1}{\ln(1 + \mu)} \ln \left(1 + \frac{\mu m}{m_p} \right) \quad 0 \leq \frac{m}{m_p} \leq 1 \quad (6.35a)$$

- The A-law for positive amplitudes is given by

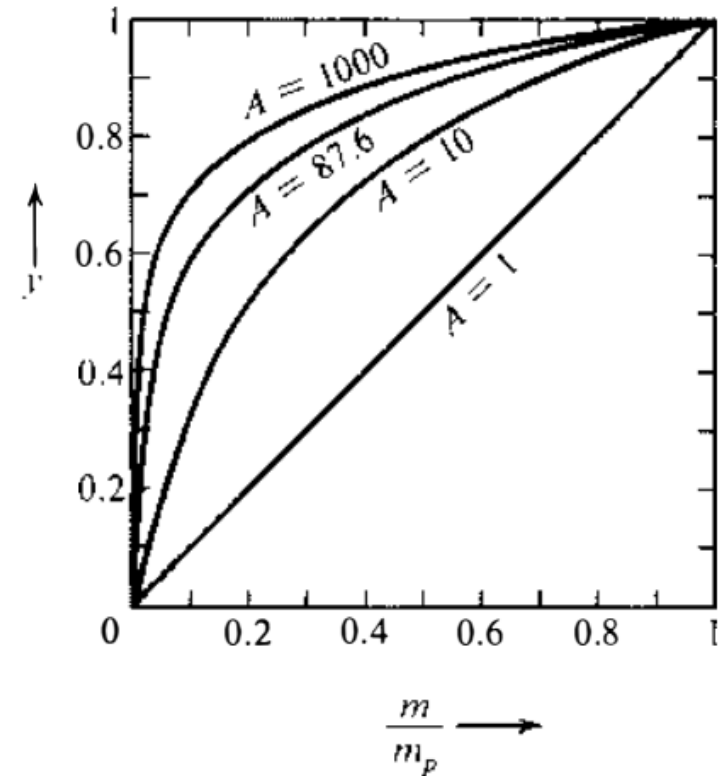
$$y = \begin{cases} \frac{A}{1 + \ln A} \left(\frac{m}{m_p} \right) & 0 \leq \frac{m}{m_p} \leq \frac{1}{A} \\ \frac{1}{1 + \ln A} \left(1 + \ln \frac{A m}{m_p} \right) & \frac{1}{A} \leq \frac{m}{m_p} \leq 1 \end{cases} \quad (6.35b)$$

These characteristics are shown in Fig. 6. 16.

Fig 6.16
(a) μ -Law
characteristic.
(b) A-Law
characteristic.



(a)



(b)

- The compression parameter μ (or A) determines the degree of compression. To obtain a nearly constant $\text{SNR} = S_o/N_o$ over a dynamic range of for input signal power 40dB, μ should be greater than 100. An optimum value of $\mu=255$ has been used for all North American 8-bit (256-level) digital terminals.
- For the A-law, a value of $A = 87.6$ gives comparable results and has been standardized by the ITU- T.
- The compressed samples must be restored to their original values at the receiver by using an **expander** with a characteristic complementary to that of the compressor. The compressor and the expander together are called the **compandor**. Figure 6.17 describes the use of compressor and expander along with a uniform quantizer to achieve nonuniform quantization.

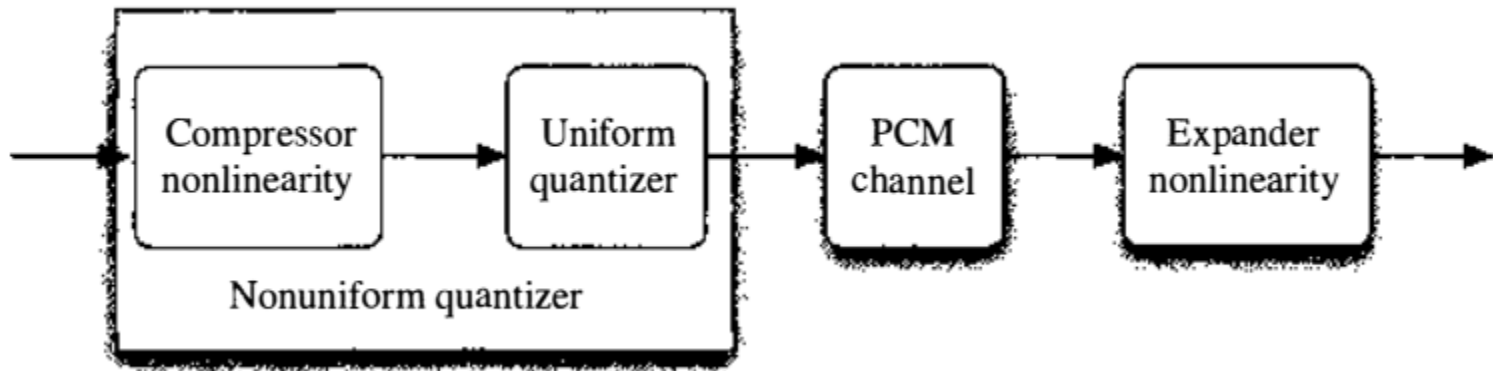


Figure 6.17 Utilization of compressor and expander for nonuniform quantization.

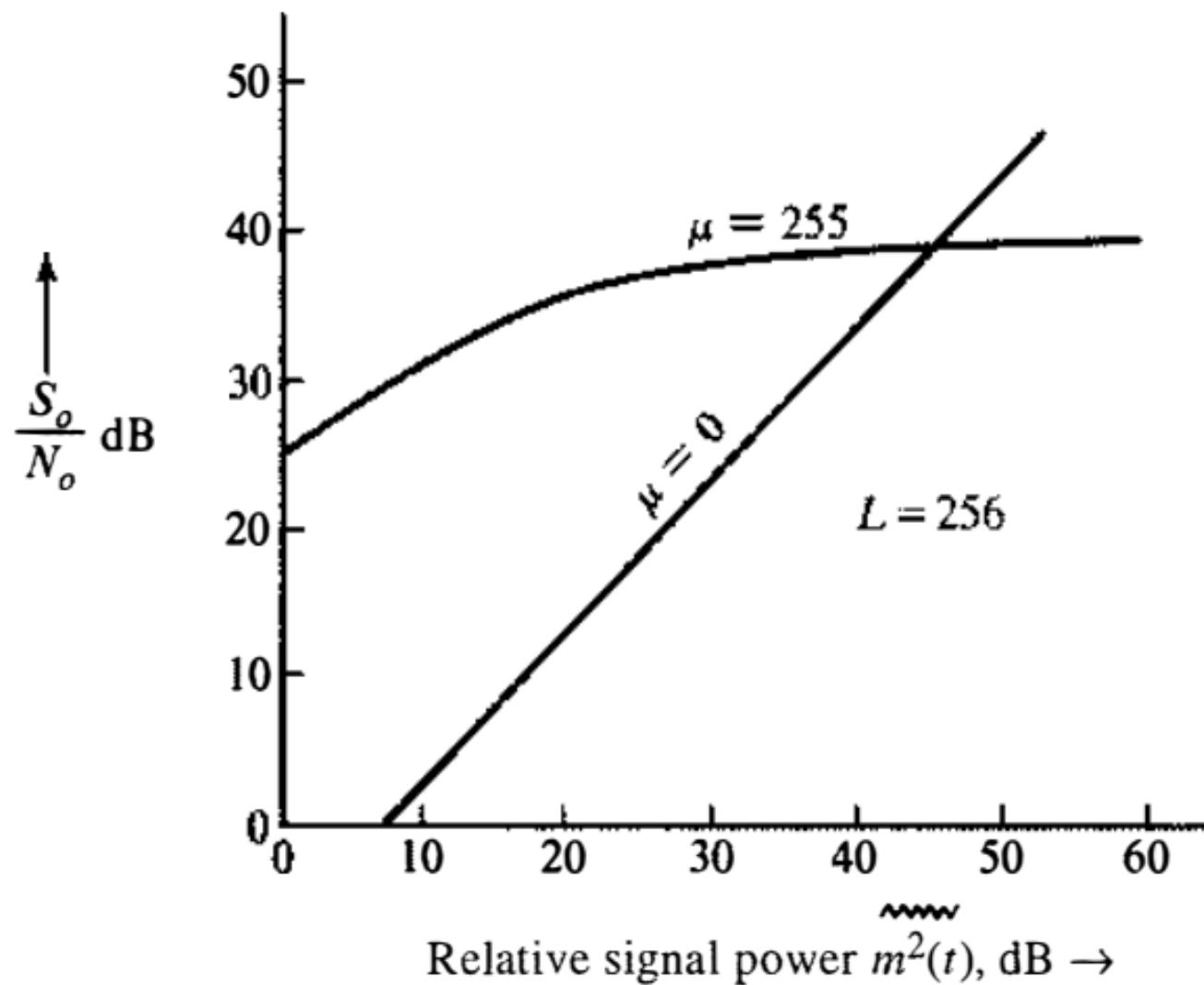
For the μ -law compandor, the output SNR is given by:

$$\frac{S_o}{N_o} \simeq \frac{3L^2}{[\ln(1 + \mu)]^2} \quad \mu^2 \gg \frac{m_p^2}{m^2(t)} \quad (6.36)$$

The output SNR for the cases of $\mu = 255$ and $\mu = 0$ (uniform quantization) as a function of the message signal power $m^2(t)$, is shown in Fig. 6.18.

Figure 6.18

Ratio of signal to
quantization
noise in PCM
with and without
compression.



6.2.4 Transmission Bandwidth and the Output SNR

- As mentioned, each one of the L quantization levels is coded by $n = \lceil \log_2(L) \rceil$ bits.
- Because a signal $m(t)$ band-limited to B Hz requires a minimum of $2B$ samples per second, we require a total of $C = 2Bn$ pulses per second, that is, $2Bn$ pieces of information per second.
- From the results of the sampling theory, we can derive one of the basic relationships in communication, which states that: a maximum of $2B$ independent pieces of information per second can be transmitted, error free, over a noiseless channel of bandwidth B Hz.
- So, the minimum bandwidth B_T needed to transmit $C = 2Bn$ pieces of information per second is given by:

$$B_T = C/2 = 2Bn/2 = Bn \quad \text{Hz}$$

Example 6.2

A signal $m(t)$ band-limited to 3 kHz is sampled at a rate $33\frac{1}{3}\%$ higher than the Nyquist rate. The maximum acceptable error in the sample amplitude (the maximum quantization error) is 0.5% of the peak amplitude m_p . The quantized samples are binary coded. Find the minimum bandwidth of a channel required to transmit the encoded binary signal. If 24 such signals are time-division-multiplexed, determine the minimum transmission bandwidth required to transmit the multiplexed signal.

The Nyquist sampling rate is $R_N = 2 \times 3000 = 6000$ Hz (samples per second). The actual sampling rate is $R_A = 6000 \times (1\frac{1}{3}) = 8000$ Hz.

The quantization step is Δv , and the maximum quantization error is $\pm\Delta v/2$.

Therefore, from Eq. (6.31),

$$\frac{\Delta v}{2} = \frac{m_p}{L} = \frac{0.5}{100} m_p \implies L = 200$$

- From Eq. (6.37), we need $n = \lceil \log_2(L) \rceil = \lceil \log_2(200) \rceil = 8$ bits per sample. Thus we require to transmit a total of $C = 8 \times 8000 = 64,000$ bits/sec. Because we can transmit up to 2 bit/s per hertz of bandwidth, we require a minimum transmission bandwidth $B_T = C/2 = 32$ kHz for each channel.
- The multiplexed signal has a total of $C_M = 24 \times 64,000 = 1.536$ Mbit/s, which requires a minimum of $1.536/2 = 0.768$ MHz = 768 kHz of transmission bandwidth.

➤ **The Relation between the Bandwidth and the Output SNR for PCM**

From Eq. (6.37), $L = 2^n$, so $L^2 = 2^{2n}$, and the output SNR in Eq. (6.34) or Eq. (6.36) can be expressed as

$$\frac{S_o}{N_o} = c(2)^{2n} \quad (6.39)$$

where

$$c = \begin{cases} \frac{3 \overline{m^2(t)}}{m_p^2} & \text{[uncompressed case, in Eq. (6.34)]} \\ \frac{3}{[\ln(1 + \mu)]^2} & \text{[compressed case, in Eq. (6.36)]} \end{cases}$$

Substitution of Eq. (6.38) into Eq. (6.39) yields

$$\frac{S_o}{N_o} = c(2)^{2B_T/B} \quad (6.40)$$

From Eq. (6.40) we observe that the SNR increases exponentially with the transmission bandwidth B_T . A small increase in bandwidth yields a large benefit in terms of SNR. This relationship is clearly seen by using the decibel scale to rewrite Eq. (6.39) as

$$\begin{aligned}
\left(\frac{S_o}{N_o}\right)_{\text{dB}} &= 10\log_{10}\left(\frac{S_o}{N_o}\right) \\
&= 10\log_{10}[c(2)^{2n}] \\
&= 10\log_{10}(c) + 20n\log_{10}(2) \\
&= (\alpha + 6n) \text{ dB}
\end{aligned} \tag{6.41}$$

where $\alpha = 10 \log_{10}(c)$. This shows that increasing n by 1 (increasing one bit in the codeword) quadruples the output SNR (a 6 dB increase). Thus, if we increase n from 8 to 9, the SNR quadruples, but the transmission bandwidth B_T increases only from $(8 \cdot 4) = 32$ kHz to $(9 \cdot 4) = 36$ kHz (an increase of only $[36 - 32/32] \cdot 100\% = 4 \cdot 100/32\% = 12.5\%$). This shows that in PCM, SNR can be controlled by controlling the transmission bandwidth.

Example 6.3: A signal $m(t)$ of bandwidth $B = 4$ kHz is transmitted using a binary companded PCM with $\mu = 100$. Compare the case of $L = 64$ with the case of $L = 256$ from the point of view of transmission bandwidth and the output SNR.

For $L = 64$, $n = 6$, and the transmission bandwidth is $nB = 24$ kHz,

$$\frac{S_o}{N_o} = (\alpha + 36) \text{ dB}$$

$$\alpha = 10 \log \frac{3}{[\ln(101)]^2} = -8.51$$

Hence,

$$\frac{S_o}{N_o} = 27.49 \text{ dB} = 561$$

For $L = 256$, $n = 8$, and the transmission bandwidth is 32 kHz,

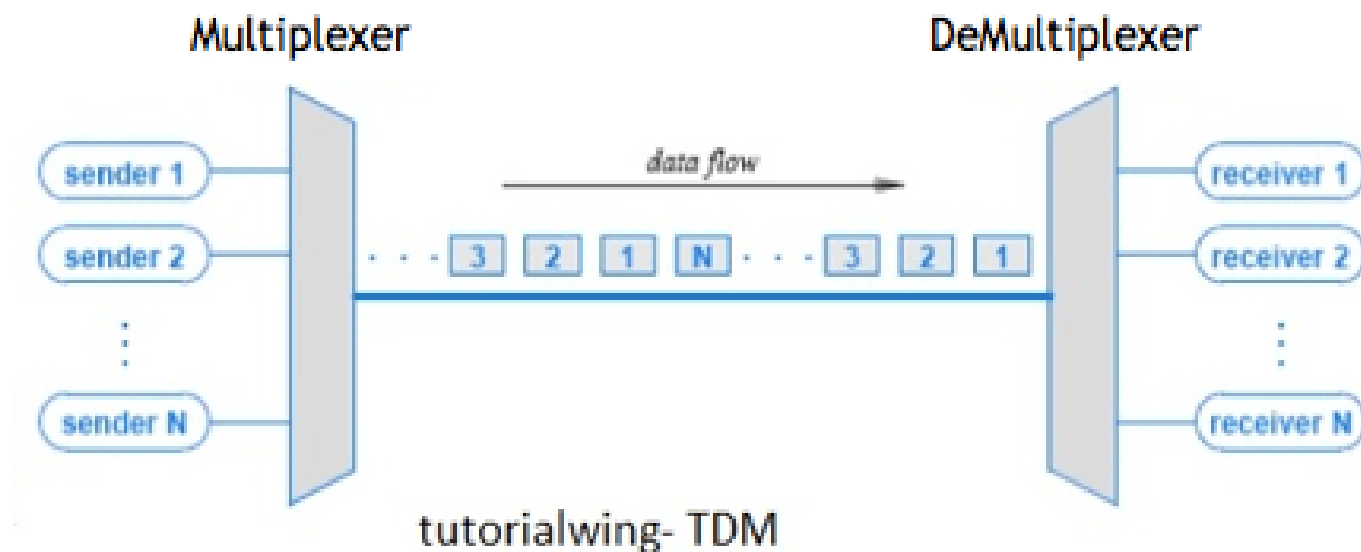
$$\frac{S_o}{N_o} = \alpha + 6n = 39.49 \text{ dB} = 8892$$

The difference between the two SNRs is 12 dB, which is a ratio of 16 ($8892/561 \approx 16$). Thus, the SNR for $L = 256$ is 16 times the SNR for $L = 64$ while the increase in bandwidth is just about 33%.

Time Division Multiplexing (TDM)

- A signal $m_1(t)$ sampled at a sampling frequency of f_s samples/sec, means that one sample from $m_1(t)$ is sent during the sampling interval $T_s = 1/f_s$ sec. For example if $f_s = 8000$ samples/sec, then one sample is sent during every $T_s = 1/8000 = 10^6/8000 = 125 \mu\text{sec}$.
- If the sample width $\ll T_s$, we can add a sample from a second signal $m_2(t)$ after the sample of $m_1(t)$ within the sampling interval T_s .

- So, two signals that are time separated can be sent over the same channel. This technique is called Time Division Multiplexing (TDM).
- Notice that in this case, two samples (from $m_1(t)$ and $m_2(t)$) are sent during time interval T_s . So the total sampling rate will be $2/T_s = 2f_s$. That is the sampling rate is doubled.
- In the same way, for N TDM signals, the final sampling rate = Nf_s samples/sec and the time between samples = $1/Nf_s$.



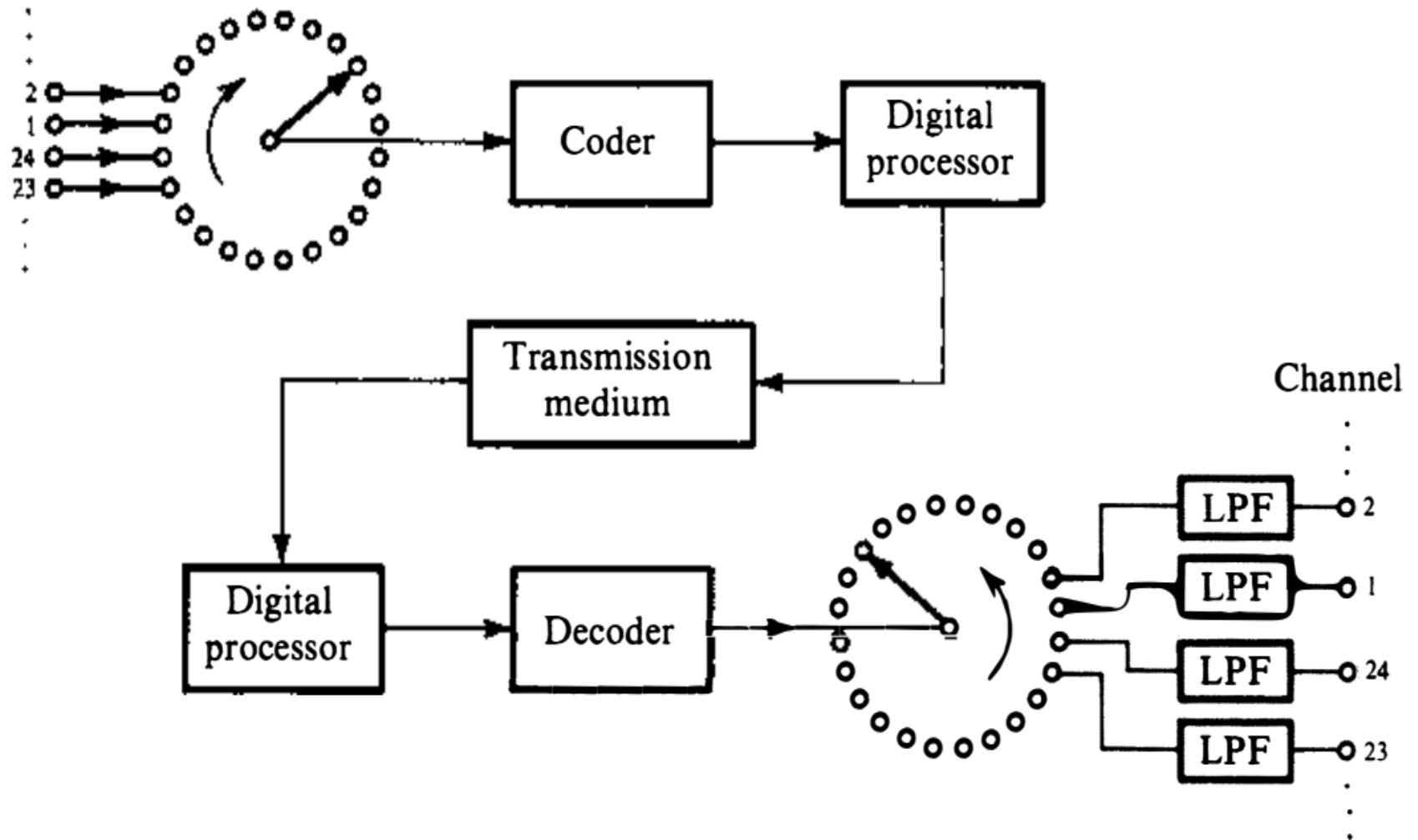
6.3 DIGITAL TELEPHONY: PCM IN T1 CARRIER SYSTEMS

➤ T1 Time Division Multiplexing

A schematic of a T1 carrier system is shown in Fig. 6.20a.

- All 24 channels are sampled in a sequence. The sampler output represents a time-division-multiplexed PAM signal.
- The multiplexed PAM signal is now applied to the input of an encoder that quantizes each sample and encodes it into 8 bits codeword (see Fig. 6.20b).
- The signal now converted to digital form, is sent over the transmission medium.
- Regenerative repeaters spaced approximately 6000 feet apart detect the pulses and retransmit new pulses.
- At the receiver, the decoder converts the binary pulses into samples (decoding).

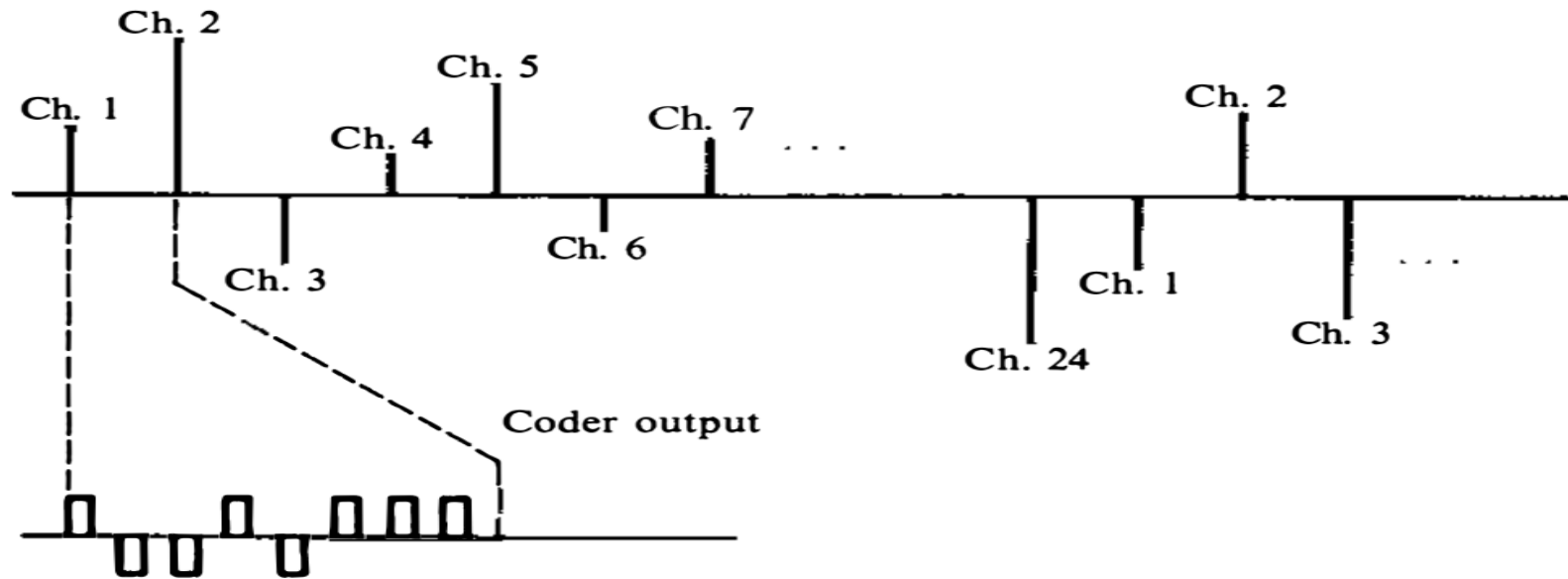
Channel



(a)

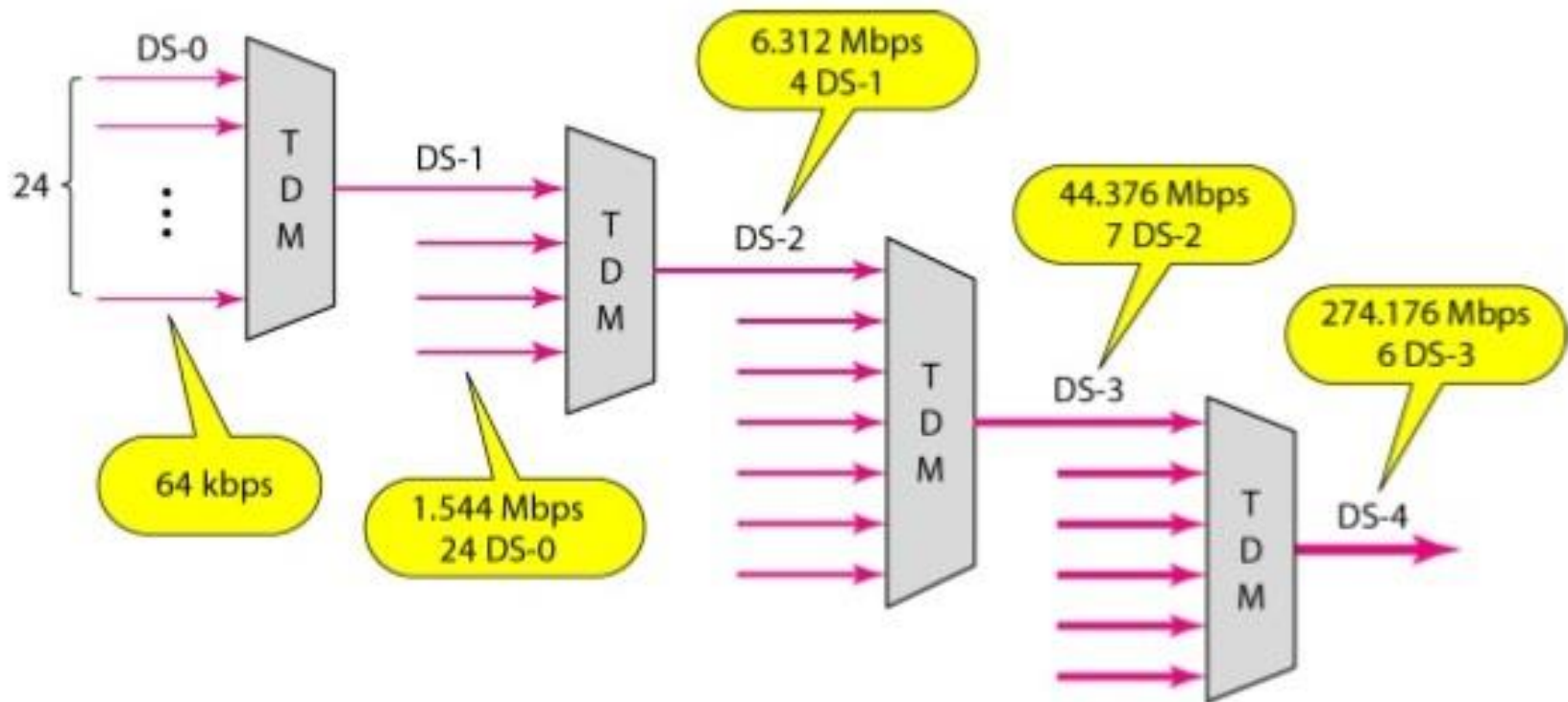
Fig. 6.20 T1 Carrier System

The samples are then demultiplexed (i.e., distributed to each of the 24 channels). The desired audio signal is reconstructed by passing the samples through a low-pass filter in each channel.



(b)

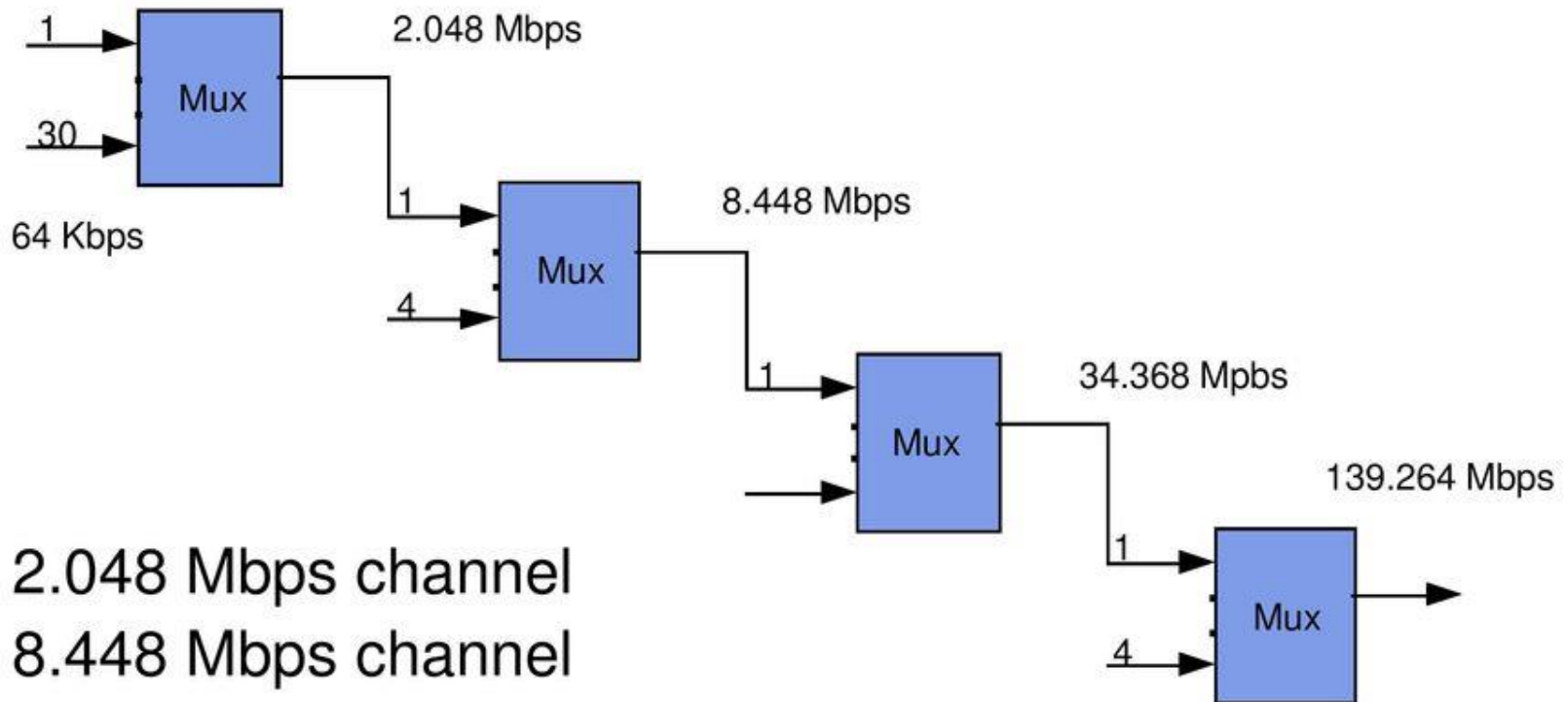
- The T1 system is called **digital signal level 1 (DS1)**.
- It operates at $64 \text{ kbits/s} \times 24 = 1.544 \text{ Mbit/s}$.
- It is used further to multiplex into progressively higher level signals DS2, DS3, and DS4.



North American digital hierarchy system

ITU-T European digital hierarchy system

It is used in Iraq



- E1, 2.048 Mbps channel
- E2, 8.448 Mbps channel
- E3, 34.368 Mbps channel
- E4, 139.264 Mbps channel

Synchronizing

- In T1 (DS-0 level), Binary codewords corresponding to samples of each of the 24 channels are multiplexed in a sequence. A segment containing one codeword (corresponding to one sample) from each of the 24 channels is called a **frame**.
- Each frame has $24 \times 8 = 192$ information bits. Because the sampling rate is 8000 samples per second, each frame takes $125 \mu s$.
- To separate information bits correctly at the receiver, it is necessary to be sure where each frame begins. Therefore, a **synchronization codeword** is added at the beginning of each frame as shown.

- The **synchronization codeword** is chosen so that it forms a special bit pattern that is unlikely to be formed in a speech signal.

